Assignment 3

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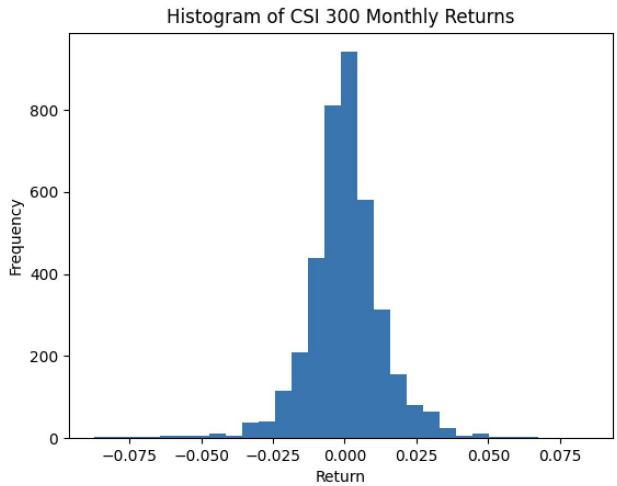
Data:

China Stock Market Series – Market Index – Trading Data of Domestic Index - Daily Market Information of Domestic Index - Closing Index[Idxtrd05]

China Stock Market Series – Stock Trading – Individual Stock Trading - Weekly Stock Price & Returns - Weekly Return Without Cash Dividend Reinvested[Wretnd] (All A shares)

Problem 1

Mean	Std	Skewness	Kurtosis
0.000163	0.01364	-0.432701	5.089395



The monthly returns of the CSI 300 index is NOT normally distributed. From the figure, the skewness is close to zero. However, the true skewness is -0.432, indicating a negative skewness. The Kurtosis is 5.0893, which far away from normal distribution's Kurtosis at 3. The mean and standard deviation are not contradict normality, but the Skewness and Kurtosis are strongly against the normality.

Problem 2

ii. Follow the instruction, I calculated the parameters and sorted by β, then selected for 10 random stocks.

Table1 Time-series regression results for first period of individual stocks

Stkcd	α	β	t_α	t_β	p_α	p_β	R^2
2260	-0.099328	-5.60822	-1.111346	-1.57801	0.316982	0.175393	0.332454
603179	0.009545	0.806237	1.460991	3.307085	0.146082	0.001177	0.067123
600275	-0.009943	1.004385	-1.228876	3.263797	0.221484	0.001426	0.080303
2776	-0.004327	1.198257	-0.636874	4.759011	0.525225	5e-06	0.136725
1256	0.077235	5.449533	0.423268	0.890117	0.745096	0.536969	0.44206
2521	-0.002421	0.960994	-0.989249	10.522892	0.324113	<2e-16	0.421462
600557	0.001573	0.630635	0.354538	3.808633	0.723427	0.000202	0.087118
2816	-0.004271	0.952878	-0.973275	5.820114	0.331983	<2e-16	0.184223
603136	0.003302	0.657109	0.624771	3.332241	0.533058	0.001082	0.068078
601816	-0.002595	0.534778	-1.109965	6.168363	0.268802	<2e-16	0.203416

iii. There are 65 portfolios in total

Table2 Time-series regression results for first period of portfolio stocks

group	α	β	t_α	t_β	p_α	p_β	R^2
1	-0.001299	1.121499	-1.323892	30.593851	0.187526	<2e-16	0.860292
2	-0.000177	0.98793	-0.234672	35.100425	0.814779	<2e-16	0.890177
3	9.5e-05	0.98972	0.118748	33.161558	0.905631	<2e-16	0.878564
4	-0.000289	1.049962	-0.354976	34.529912	0.7231	<2e-16	0.886931
5	-0.000959	1.047675	-0.95601	27.953159	0.340585	<2e-16	0.837151
6	0.000825	0.982398	0.935432	29.808498	0.351049	<2e-16	0.853923
7	0.000826	1.117134	0.888304	32.164419	0.375781	<2e-16	0.871898
8	-0.000314	1.030009	-0.41448	36.350591	0.679107	<2e-16	0.896835
9	0.000327	1.002959	0.32365	26.556839	0.746648	<2e-16	0.822692
10	-0.001786	1.043977	-0.980523	15.34661	0.328386	<2e-16	0.607761
11	0.000585	0.959797	0.645879	28.349527	0.519332	<2e-16	0.840954
12	0.000146	1.062319	0.195064	37.945131	0.845603	<2e-16	0.904513
13	-6.90E-05	1.078663	-0.090456	37.988393	0.928044	<2e-16	0.904709
14	0.000708	1.118549	0.933472	39.49735	0.352057	<2e-16	0.911217
15	-7.20E-05	0.99957	-0.106978	39.536339	0.914948	<2e-16	0.911377
16	0.000183	1.091132	0.244586	39.084921	0.807107	<2e-16	0.909504
17	0.000347	0.984624	0.486376	36.936703	0.627401	<2e-16	0.899757
18	9.7e-05	1.035607	0.110796	31.671015	0.911925	<2e-16	0.868404
19	6.2e-05	0.994285	0.074293	31.824615	0.940875	<2e-16	0.869506
20	-0.000935	1.015709	-1.065988	31.000125	0.288119	<2e-16	0.863433
21	0.000176	1.100125	0.2167	36.356201	0.828733	<2e-16	0.896863
22	0.000679	0.957826	0.986876	37.253551	0.325272	<2e-16	0.901288
23	-0.000806	1.099904	-0.989431	36.140682	0.324025	<2e-16	0.895758
24	-0.000846	1.105443	-1.120184	39.177606	0.264402	<2e-16	0.909893
25	0.000562	0.993712	0.660296	31.264475	0.510063	<2e-16	0.865423
26	-0.000336	1.029609	-0.380906	31.292669	0.703805	<2e-16	0.865633
27	0.000526	1.004955	0.609679	31.167519	0.542985	<2e-16	0.864698
28	0.000702	1.063605	0.708554	28.755855	0.479687	<2e-16	0.844724

29	9e-06	1.012958	0.009096	29.011906	0.992755	<2e-16	0.847035
30	0.001367	1.125357	0.758391	16.713642	0.449391	<2e-16	0.647615
31	0.000865	0.785553	0.693984	16.883045	0.488752	<2e-16	0.652204
32	-0.000232	0.90401	-0.277395	28.973113	0.781854	<2e-16	0.846688
33	-0.001412	1.037974	-1.802268	35.466623	0.073485	<2e-16	0.89219
34	0.000461	0.950601	0.594448	32.841395	0.553096	<2e-16	0.876479
35	-7.60E-05	0.982762	-0.104118	35.965246	0.917213	<2e-16	0.894846
36	-0.000749	0.965162	-1.037483	35.776677	0.301158	<2e-16	0.893853
37	0.000153	1.02324	0.218438	39.1672	0.827381	<2e-16	0.90985
38	0.000797	1.012734	1.051202	35.755892	0.294834	<2e-16	0.893742
39	0.000165	0.93289	0.229202	34.790381	0.81902	<2e-16	0.88843
40	0.000245	0.959928	0.366065	38.405857	0.714826	<2e-16	0.906577
41	-0.000903	0.943268	-1.240192	34.693915	0.216815	<2e-16	0.887878
42	-0.001675	0.934966	-2.334616	34.885115	0.020872	<2e-16	0.888968
43	-0.000627	0.991427	-0.810823	34.299807	0.418735	<2e-16	0.885583
44	0.000467	0.958317	0.602131	33.10267	0.547984	<2e-16	0.878184
45	-0.00087	0.951088	-1.097097	32.101529	0.274334	<2e-16	0.87146
46	0.000233	0.885337	0.304614	31.054839	0.761077	<2e-16	0.863848
47	0.00011	0.97147	0.132985	31.464351	0.894381	<2e-16	0.866901
48	0.000443	0.886059	0.46249	24.746554	0.644391	<2e-16	0.801149
49	-0.001165	0.871601	-1.261318	25.27292	0.209127	<2e-16	0.80777
50	-0.000111	0.951401	-0.113212	26.057774	0.910012	<2e-16	0.81709
51	-0.000339	0.850138	-0.330458	22.188389	0.741509	<2e-16	0.764094
52	0.000185	0.996911	0.275901	39.849737	0.782999	<2e-16	0.912644
53	-0.00016	1.066802	-0.194658	34.667822	0.845921	<2e-16	0.887728
54	0.000432	1.057071	0.492228	32.259325	0.623269	<2e-16	0.872554
55	0.001236	0.947811	1.1043	22.678373	0.271208	<2e-16	0.771877
56	0.001202	0.950635	1.188298	25.169018	0.236569	<2e-16	0.806488
57	0.001151	0.97723	1.147539	26.086615	0.252962	<2e-16	0.81742
58	-0.000216	0.923998	-0.261506	29.933109	0.794056	<2e-16	0.854961
59	-7.00E-06	1.058478	-0.008492	33.048671	0.993236	<2e-16	0.877835
60	0.000101	1.008858	0.104653	28.004143	0.916789	<2e-16	0.837647
61	0.000245	0.94394	0.311266	32.156429	0.756025	<2e-16	0.871842
62	0.000811	1.061635	0.90189	31.615219	0.368542	<2e-16	0.868001
63	0.000249	1.020647	0.226604	24.846445	0.821036	<2e-16	0.80243
64	0.009037	1.250284	1.576811	5.51851	0.117251	<2e-16	0.188623
65	0.003657	0.883585	1.158265	6.679868	0.249037	<2e-16	0.269415

iv. The two periods have same number of weeks.

Table3 Cross-sectional regression results for second period of portfolio stocks

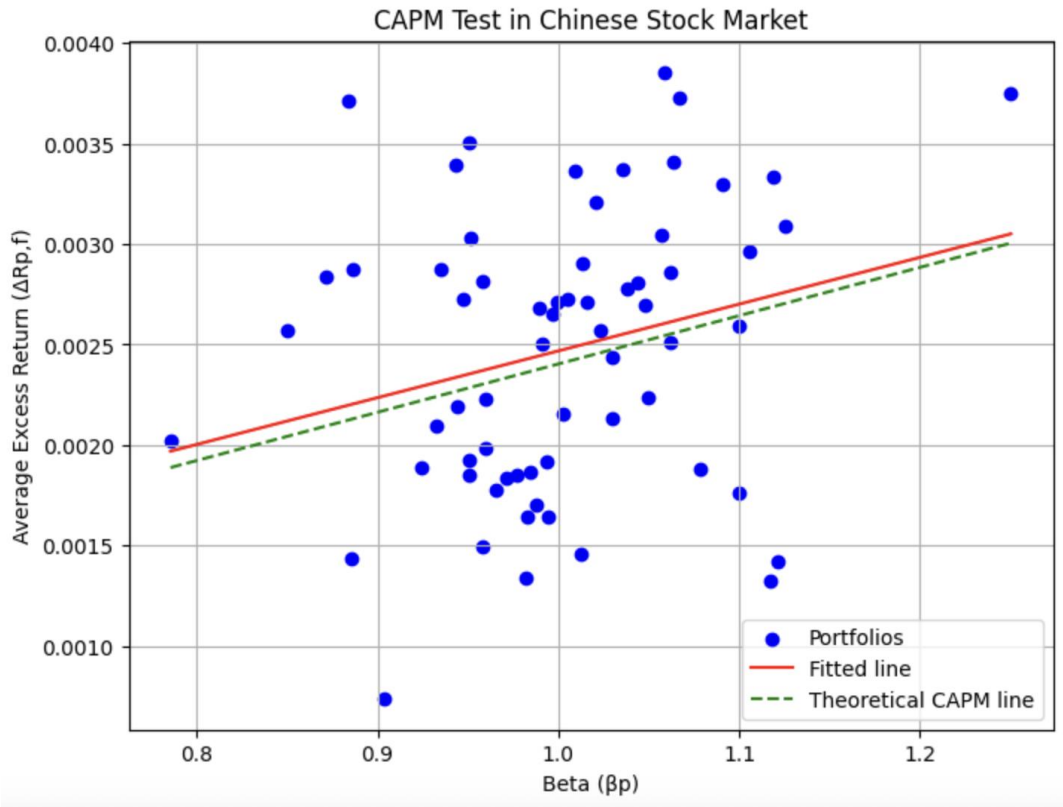
γ_0	γ_1	t_γ_0	t_γ_1	p_γ_0	p_γ_1	R^2
0.000141	0.002327	0.124748	2.066447	0.90112	0.042903	0.063478

The R^2 value is 0.063478, indicating a low goodness of fit; $\gamma_1 = 0.002327$. The t-statistic for γ_1 reveals a significant positive correlation between returns and systematic risk, suggesting that

returns increase as risk increases -- a finding consistent with the CAPM model. Conversely, γ_0 is not statistically different from zero, which is in line with the CAPM implication that no abnormal returns should exist, indicating that other factors may also influence asset pricing.

v.

Figure1



The empirical results indicate that there is a positive linear relationship between portfolio betas and their average excess returns, which is broadly consistent with the predictions of the CAPM. However, the dispersion of the data points around the fitted regression line suggests that beta has limited explanatory power for returns. In addition, the fitted line does not perfectly coincide with the theoretical CAPM line, indicating deviations from the model.

Therefore, while systematic risk plays an important role in asset pricing, it is not the sole determinant. Other factors, such as firm-specific characteristics or additional risk factors, may also influence expected returns in the Chinese stock market.